Multi-mode wave ports in EMerge

Port eigenproblem, evanescent modes, mode tracking, the propagation constant over frequency, and Floquet ports

1 Overview

A `WavePort` terminates a guiding structure with a fixed set of N port modes. Each mode m is absorbed and excited with its own propagation constant $\gamma_m = j\beta_m$, which is what a port needs when its modes have different β (the even and odd mode of coupled lines, TE_{10} and TE_{30} of a waveguide). The implementation rests on four pieces:

1. the port eigenproblem, solved at every frequency, **including evanescent modes** (Section 2);
2. a normalization and a projection that work for propagating and evanescent modes alike (Section 3);
3. the dense wave port boundary condition (WPBC), one rank-1 term per mode (Section 4);
4. a **fixed mode set**: the modes are chosen at the highest frequency and followed down in frequency by a predictor (an extrapolation of β^2) and a field-overlap match (Section 5, Section 6).

The predicted β^2 is **only** used to steer the eigen solver and to identify the modes. The β_m and the mode fields that enter the system matrix are always the eigen solution at that frequency, so the accuracy of the extrapolation never limits the accuracy of the S-parameters. It only has to be good enough to pick the right eigenvalue (Section 6.3).

The `FloquetPort` of periodic cells uses the same ideas with analytic modes: TE and TM Floquet harmonics, including grating modes, each absorbed with its own γ by a dense projection term (Section 8).

2 The port eigenproblem

On the port face, with fields $\boldsymbol{E} = (\boldsymbol{E}_t + \hat{z}E_z)e^{-j\beta z}$ ($\hat{z}$ points into the model), EMerge solves the Nédélec–Lagrange formulation in the scaled unknowns

$$\boldsymbol{e}_t = \beta \boldsymbol{E}_t, \quad e_z = -jE_z, \tag{1}$$

which turns Maxwell's equations into the generalized eigenproblem

$$\boldsymbol{A}(k_0)\boldsymbol{x} = \lambda \boldsymbol{B}(k_0)\boldsymbol{x}, \quad \lambda = -\beta^2. \tag{2}$$

eigenvalue	meaning	treatment
$\lambda < 0$ ($\beta^2 > 0$)	propagating mode	kept
$\lambda > 0$ ($\beta^2 < 0$)	evanescent mode, $\beta = -j\alpha$	kept (new)
$\lambda \approx 0$	spurious gradient (null-space) solutions	removed: $\lvert\beta^2\rvert < 10^{-4} k_0^2 \varepsilon_{r,\max}\mu_{r,\max}$
large residual	unconverged Arnoldi vector	removed: $\lVert \boldsymbol{A}\boldsymbol{x} - \lambda\boldsymbol{B}\boldsymbol{x}\rVert > 10^{-6}(\cdots)$

Table 1: The spectrum of Equation 2. The spurious cluster sits at $\beta^2 = 0$ to round-off (53 of them on a 10×10.16 mm port at 8 GHz), while the physical evanescent modes sit where $\beta^2 = k_0^2 - k_c^2$ puts them (TE_{01} -67440 vs. -67520 analytic, TE_{10} -70521 vs. -70606). The two groups are cleanly separated.

The old solvers (`eig_boundary`) only returned $\beta^2 \geq 1$, so evanescent modes never appeared. The new routine `SolveRoutine.eig_boundary_shifts` takes a list of target values of β^2 ("shifts"): small ports (< 500 DoFs) are solved densely, larger ones by shift-invert Arnoldi, one factorization per cluster of shifts. A shift must not sit on the spurious cluster at 0; since the shifts are predictions of physical β^2 values, they don't, except for a mode that is exactly at cutoff at a sample frequency (a known limitation, Section 9).

Branch of $\boldsymbol{\beta}$. $\beta = \sqrt{-\lambda}$ is taken with $\mathrm{Im}\ \beta \leq 0$. Then an evanescent mode $e^{-j\beta z} = e^{-\alpha z}$ decays along its direction of travel, like a lossy mode, and the outgoing (reflected) evanescent wave $e^{+j\beta z}$ decays away from the model, into the port.

3 Normalization and modal orthogonality

Every `WavePort` mode is scaled to

$$\left| \frac{1}{2} \iint_S (\boldsymbol{e}_m \times \boldsymbol{h}_m) \cdot \hat{n} \, \mathrm{d}S \right| = 1 \tag{3}$$

without a complex conjugate. For a lossless propagating mode with real transverse fields this is its power, so the usual 1 W normalization is unchanged. An evanescent mode carries no power ($\mathrm{Re} \int (\boldsymbol{e} \times \boldsymbol{h}^*) \cdot \hat{n} = 0$), but Equation 3 is still finite, so it can be normalized at all. Equation 3 is also the natural normalization because of the **reciprocity (biorthogonality) relation**

$$\iint_S (\boldsymbol{e}_m \times \boldsymbol{h}_n) \cdot \hat{n} \, \mathrm{d}S = 0, \quad m \neq n, \tag{4}$$

which follows from Lorentz reciprocity for any reciprocal (also lossy) cross-section, and holds for propagating and evanescent modes alike. The conjugated power product is only orthogonal for lossless propagating modes.

4 Wave port boundary condition and S-parameters

With the mode profile $\boldsymbol{w}_m = \gamma_m \boldsymbol{e}_{tm} + \nabla_t e_{zm}$, which is proportional to $\boldsymbol{h}_{tm} \times \hat{n}$, the port operator reads

$$\hat{n} \times \nabla \times \boldsymbol{E} = \sum_{m=1}^{N} \boldsymbol{w}_m \frac{\int \boldsymbol{E}_t \cdot \boldsymbol{w}_m \, \mathrm{d}S}{\int \boldsymbol{e}_{tm} \cdot \boldsymbol{w}_m \, \mathrm{d}S} - 2a_m \boldsymbol{w}_m. \tag{5}$$

Because of Equation 4, $\int \boldsymbol{e}_{tn} \cdot \boldsymbol{w}_m \, \mathrm{d}S = 0$ for $n \neq m$: the sum is a projection on the port modes, each mode is absorbed with its own γ_m, and the incident amplitude a_m excites exactly mode m. In the FEM, Equation 5 adds one dense rank-1 block per mode,

$$-\frac{1}{j\omega\mu_0\kappa_m} \boldsymbol{G}_m \boldsymbol{G}_m^T, \quad \boldsymbol{G}_m = \iint \boldsymbol{w}_m \cdot \boldsymbol{N}_i \, \mathrm{d}S, \quad \kappa_m = \frac{j}{\omega\mu_0} \iint \boldsymbol{e}_{tm} \cdot \boldsymbol{w}_m \, \mathrm{d}S, \tag{6}$$

and the excitation vector $-2\boldsymbol{G}_m$. For an evanescent mode $\gamma_m = \alpha_m$ is real: the term then absorbs the exponential tail of a mode that a discontinuity generates near the port, so a port can sit close to the discontinuity (Section 7). The outgoing amplitude of mode m is the same projection,

$$b_m = \frac{\int (\boldsymbol{E} - a_m \boldsymbol{e}_m) \cdot \boldsymbol{w}_m \, \mathrm{d}S}{\int \boldsymbol{e}_{tm} \cdot \boldsymbol{w}_m \, \mathrm{d}S}, \quad S_{mn} = \frac{b_m}{a_n}. \tag{7}$$

Between propagating modes $|S_{mn}|^2$ is a power ratio. Entries that involve an evanescent mode are amplitude ratios in the normalization Equation 3 and may exceed 1; for that reason EMerge's power caps (`mw_cap_sp_col`, `mw_cap_sp_single`) now only act on the propagating block.

5 A fixed set of modes

Selection. The modes are fixed at the highest frequency $f_{\max}$: the N modes with the largest $\mathrm{Re}\ \beta^2$ there, or, with `number_of_modes=None`, all modes that propagate at $f_{\max}$. Evanescent modes are included when fewer than N modes propagate. If modes N and $N+1$ are degenerate, a warning asks for both: with only one, the port cannot absorb a field of that β.

Tracking. From $f_{\max}$ the modes are followed through the sweep frequencies, one step at a time. At each new frequency:

1. predict β_m^2 for every tracked mode (Section 6);
2. solve Equation 2 for the modes closest to the predictions (about $2N + 4$ candidates);
3. match tracked modes to candidates with a linear assignment (Hungarian) on the cost

$$C_{mc} = -\mathrm{CMAC}_{mc} + \frac{1}{2} \frac{\left|\beta_c^2 - \beta_{\mathrm{pred},m}^2\right|}{k_0^2 \varepsilon_{r,\max} \mu_{r,\max}}, \quad \mathrm{CMAC}_{mc} = \frac{\left|\boldsymbol{u}_m^H \boldsymbol{v}_c\right|^2}{\|\boldsymbol{u}_m\|^2 \|\boldsymbol{v}_c\|^2}, \tag{8}$$

   where $\boldsymbol{u}_m$, $\boldsymbol{v}_c$ are the transverse field coefficients at the previous and the current frequency (same mesh, same DoFs);
4. rotate the phase of each matched mode so that $\boldsymbol{u}_m^H \boldsymbol{v}_c$ is real and positive.

Step 3 keeps every S-matrix index attached to the same physical mode through cutoff, where β^2 changes sign, and through **mode crossings**, where ordering by β would swap the labels. Step 4 makes the modes, and therefore the S-parameters, smooth functions of frequency. A CMAC below 0.7 gives a warning (too coarse a frequency step).

6 Predicting β^2 over frequency

6.1 Homogeneous cross-sections

If the port face is one isotropic material, the modes are pure TE, TM or TEM with frequency-independent transverse profiles, and

$$\beta^2(k_0^2) = \varepsilon_r \mu_r k_0^2 - k_c^2. \tag{9}$$

β^2 is exactly linear in k_0^2 with slope $\varepsilon_r\mu_r$ (Figure 1). Both β and the modes at any frequency follow from one solve, which is what a homogeneous `WavePort` does (Section 6.4).

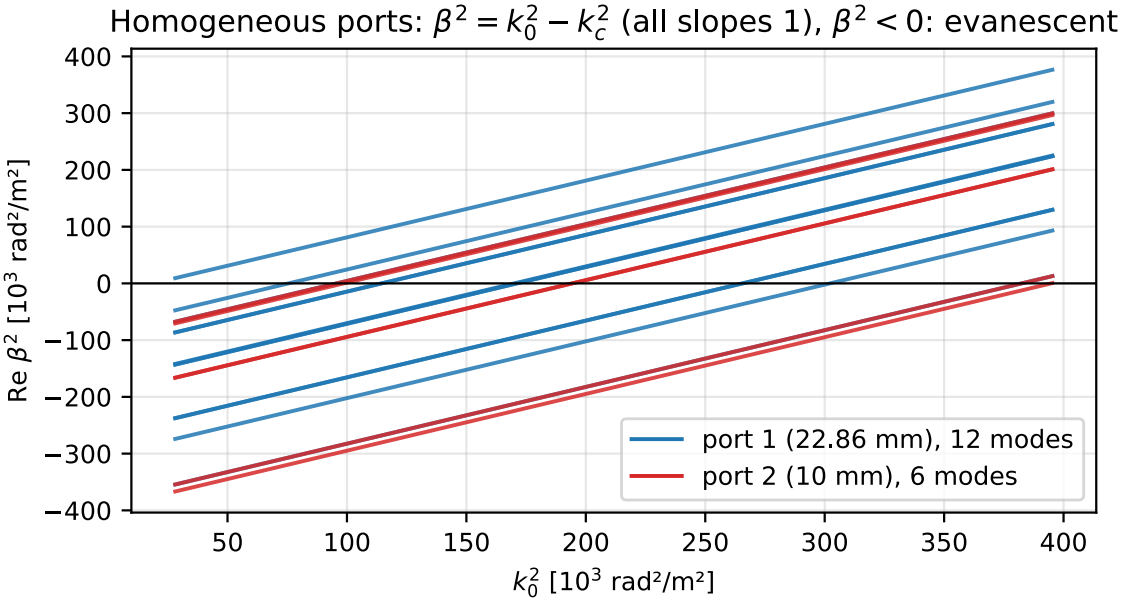


Figure 1: β^2 of the 12 + 6 tracked modes of the two homogeneous ports of the step in Section 7, 8–30 GHz. Every line is straight with slope 1; the zero crossings are the cutoffs. Below its cutoff a mode stays in the port as an evanescent mode.

This is also where the `ModalPort` bug sat: modes computed once were extrapolated with $\beta^2 = \beta_0^2 + k_0^2 - k_{0,0}^2$, which leaves out $\varepsilon_r\mu_r$, and their power normalization was not updated (TE: $P \propto \beta/\omega$). A uniform guide with $\varepsilon_r = 2.2$ reflected $|S_{11}| = 0.41$ at 6 GHz. `get_beta` now uses Equation 9, and a `ModalPort` reuses its modes only when they are exact at every frequency (one material, only TEM modes). All other ports are solved again at every frequency.

6.2 Inhomogeneous cross-sections: the slope $d\beta^2/dk_0^2$

In the scaled unknowns, the Rayleigh quotient of the curl–curl operator becomes rational in $B = \beta^2$ for a fixed field, with $K = k_0^2$:

$$K = \frac{a + Bb}{c + Bd}, \qquad \begin{cases} a = \int \mu_r^{-1} \, |\nabla_t \times \boldsymbol{e}_t|^2 & \\ & b = \int \mu_r^{-1} \, |\boldsymbol{e}_t + \nabla_t e_z|^2 \\ c = \int \varepsilon_r \, |\boldsymbol{e}_t|^2 & \\ & d = \int \varepsilon_r \, |e_z|^2 \, . \end{cases} \tag{10}$$

A mode makes $G(K, B; \boldsymbol{e}) = a + Bb - K(c + Bd)$ stationary in $\boldsymbol{e}$. Along the dispersion curve the change of the field therefore does not contribute (Hellmann–Feynman), and

$$s(K) \equiv \frac{d\beta^2}{dk_0^2} = \frac{c + \beta^2 d}{b - k_0^2 d} = n_{\text{eff}} n_g = \frac{c^2}{v_p v_g}. \tag{11}$$

(For lossy, reciprocal media the same holds with the non-conjugated bilinear forms, because the operator is complex symmetric.) Equation 11 says how fast β^2 can turn:

- **Homogeneous:** $s = \varepsilon_r \mu_r$ for every mode, Equation 9.
- **TE-type modes** ($e_z = 0$, so $d = 0$), and scalar, separable problems such as layered guides:

$$s = \frac{\int \varepsilon_r \, |\boldsymbol{e}_t|^2}{\int \mu_r^{-1} \, |\boldsymbol{e}_t|^2} \in [\min \varepsilon_r \mu_r, \max \varepsilon_r \mu_r] \tag{12}$$

  rigorously: the slope is an average of $\varepsilon_r \mu_r$, weighted with where the field lives. At low frequency the field spreads over all materials, at high frequency it concentrates in the densest one, and $s \to \max \varepsilon_r \mu_r$.
- **Lossless propagating modes:** $s > 0$, since $v_g = P/W > 0$ for a forward mode.
- **Hybrid modes** (quasi-TEM, LSE/LSM): the e_z terms in $b - k_0^2 d$ are not sign definite, and I have no proof that Equation 12 holds. In every case computed here it does (Figure 3): the slab guide stays in $[1.34, 3.94]$ against the bound $[1, 4]$, the microstrip quasi-TEM mode in $[3.31, 4.10]$ against $[1, 4.4]$, and the box modes of the microstrip in $[1.07, 3.28]$. The tracker therefore does not rely on Equation 12, but it is a useful sanity check.

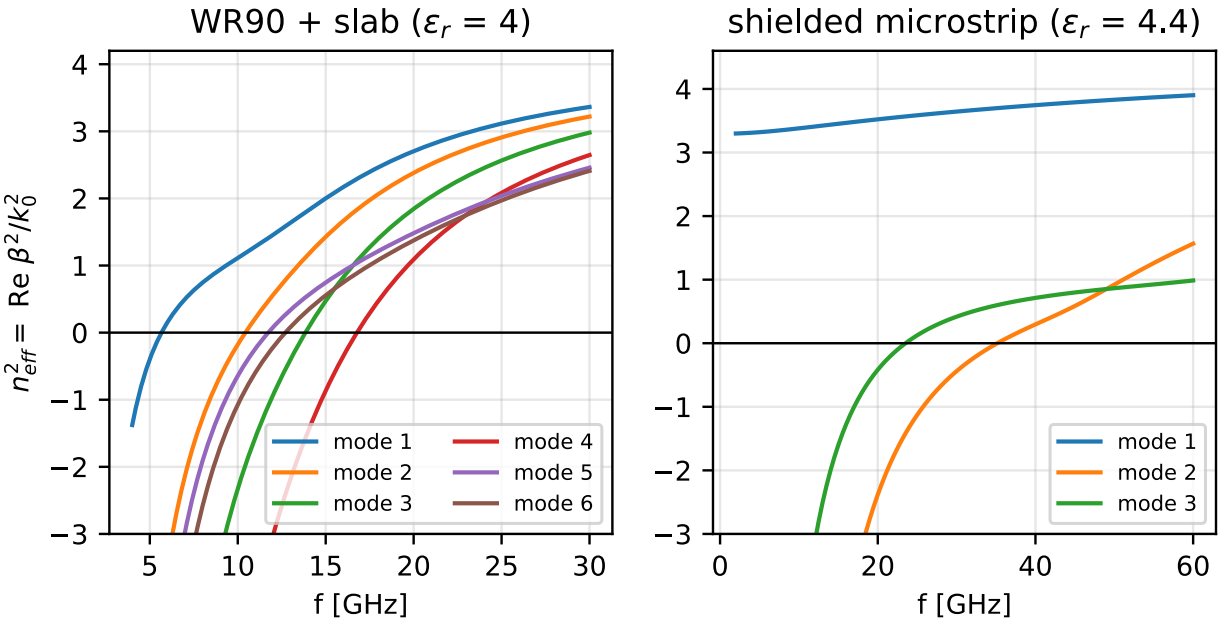


Figure 2: Tracked modes of two inhomogeneous ports (port solves only). Left: WR90 with a 3 mm slab of $\varepsilon_r = 4$ on the bottom wall; modes 4 and 5 cross near 26 GHz and keep their labels. Right: a shielded microstrip line ($\varepsilon_r = 4.4$): the quasi-TEM mode (1) and two box modes, which cross near 49 GHz.

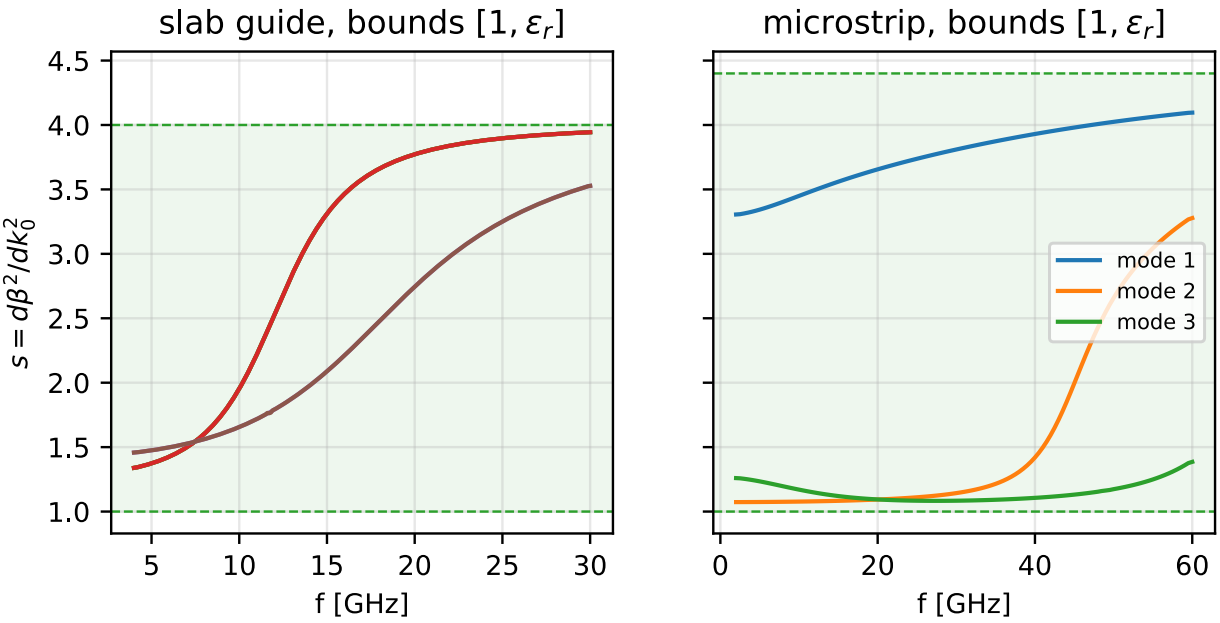


Figure 3: The slope Equation 11, from the tracked β^2, against $[\min \varepsilon_r\mu_r, \max \varepsilon_r\mu_r]$ (shaded). Modes 1–4 of the slab guide are TE_{m0}-like with the same field across the slab, so they share one slope curve (the problem separates in x and y).

6.3 The predictor and its error

With the two nearest computed frequencies K_1, K_2, the prediction at K is the secant

$$\beta^2_{\mathrm{pred}} = \beta^2(K_1) + \frac{\beta^2(K_1) - \beta^2(K_2)}{K_1 - K_2}(K - K_1), \tag{13}$$

and with one computed frequency the slope $\varepsilon_r\mu_r$ of the face (exact if homogeneous). Two error estimates follow from Equation 11:

- **Guaranteed (first order).** If $s \in [s_{\mathrm{lo}}, s_{\mathrm{hi}}]$ on the step, the true β^2 lies in a cone around the last point and

$$\left|\beta^2_{\mathrm{pred}} - \beta^2\right| \le (s_{\mathrm{hi}} - s_{\mathrm{lo}})|\Delta K|, \quad \frac{|\Delta\beta^2|}{k_0^2} \lesssim 2(s_{\mathrm{hi}} - s_{\mathrm{lo}})\frac{\Delta f}{f}. \tag{14}$$

  This holds whenever Equation 12 does, without any smoothness assumption, and is zero for homogeneous ports.
- **Typical (second order).** For a smooth s the secant error is $\approx s'(K)\Delta K^2$, so it scales as Δf^2 (Figure 4). In the slab guide at 0.1 GHz steps it is at most $5.5 \cdot 10^{-4} k_0^2$, while the guaranteed bound is $2 \cdot 3 \cdot 0.1/30 = 0.02$.

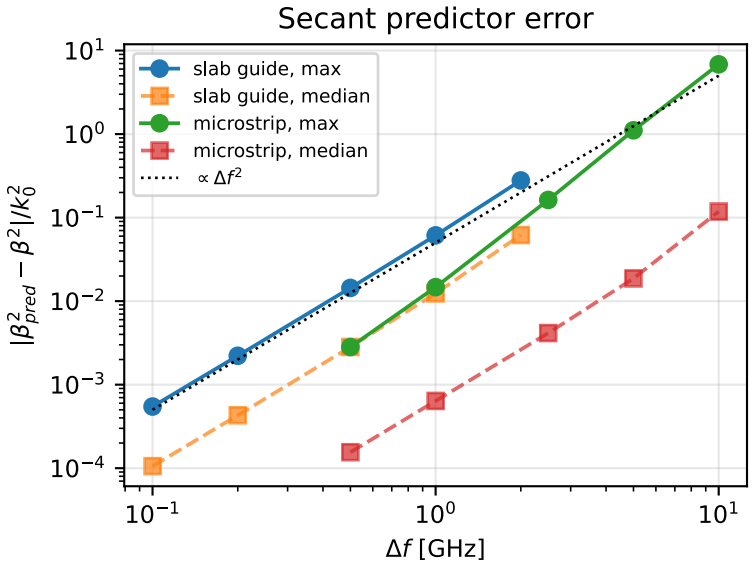


Figure 4: Measured secant error of β^2 against the frequency step, marching down in frequency as the tracker does. The slope of 2 in the log–log plot is the second-order behaviour.

How accurate does the prediction have to be? The prediction is not used in the solve: γ_m, κ_m and the mode fields in Equation 5 are the eigen solution at that frequency, to the accuracy of the 2D mesh. The prediction has two jobs:

1. **Shift of the eigen solver.** Shift-invert finds the eigenvalues closest to the shift whatever its error. The only requirement is to stay away from the spurious cluster at $\beta^2 = 0$.
2. **Identification.** The match must be unambiguous. That holds if the prediction error is below half the gap to the nearest other mode, *or* if the fields tell the modes apart. Near crossings the gap goes to zero, so there the CMAC term decides. In the slab guide the lowest CMAC over 261 frequencies is 0.999, through the crossing. In the microstrip it drops to 0.78 at the box mode crossing (the solver returns mixtures of two nearly degenerate modes) and is back to 1.000 one step later, with the labels kept.

So a coarser predictor would still give exact S-parameters, and the step size is set by the identification, not by the accuracy of β.

6.4 Homogeneous ports: one solve, scaled modes

A `WavePort` detects a homogeneous port face and then solves its modes only once, at $f_{\max}$. The port is scaled instead of solved when all of these hold:

- the face is one isotropic material,
- that material has no conductivity (σ adds $-j\sigma/(\omega\varepsilon_0)$ to ε_r, so ε_r changes with frequency),
- ε_r and μ_r are the same at $f_{\min}$ and $f_{\max}$ (no dispersion).

A constant loss tangent is allowed: it makes ε_r complex but frequency independent. The walls are PEC in the port eigenproblem at every frequency, also when they carry a surface impedance, so lossy walls do not matter here.

At another frequency β follows from Equation 9, with the branch Im $\beta \leq 0$. The fields follow from the mode type:

- TE and TEM modes keep their fields;
- a TM mode has $\boldsymbol{E}_t = -j\beta/k_c^2 \nabla_t E_z$, so its $\boldsymbol{E}_t$ is scaled by β/β_{ref};
- every mode is then normalized again (Section 3).

The scaled modes are the same fields the eigen solver would return, so the S-parameters do not change. On the step of Section 7 they agree with the per-frequency solve to $1.5 \cdot 10^{-5}$. The port mode time drops from 27–29 s to 2 s of the sweep.

Degenerate TE/TM pairs. An eigen solver returns an arbitrary basis of a degenerate pair, for example $\text{TE}_{11}/\text{TM}_{11}$ of a rectangular guide, and a mix of TE and TM cannot be scaled with one factor. Such a group is split first:

1. the TE combinations are those with $E_z = 0$ and the TM combinations those with $H_z = 0$;
2. each subspace gets a real basis (with E_z rotated by $-j\beta/\ |\beta|$, a TE or TM mode is real up to one phase);

3. modes of one type are made orthogonal in $\iint (\boldsymbol{e}_m \times \boldsymbol{h}_n) \cdot \hat{\boldsymbol{n}}\, \mathrm{d}S$.

A group whose modes are already pure keeps the basis of the solver. If a mode is neither TE, TM nor TEM, the port falls back to solving and tracking.

6.5 Why not interpolate inhomogeneous ports?

For inhomogeneous ports an interpolated β would also need interpolated mode fields. That requires a consistent gauge and a treatment of degenerate subspaces, and gives up the exactness above for an error that has to be controlled. The re-solve is cheap: a port solve costs 0.2–0.5 s per frequency.

7 Validation

H-plane step from WR90 (22.86 mm) to a 10 mm wide guide, both 10.16 mm high, 8–30 GHz, 45 frequencies, ports with 12 and 6 modes (all modes that propagate at 30 GHz). The reference is a mode-matching solution of the same step (TE_{n0} modes, 80 + 35 modes, converged to $4 \cdot 10^{-4}$), de-embedded to the junction. Below 15.0 GHz **no** mode propagates at the narrow port: TE_{10} there is evanescent and $|S_{11}| = 1$.

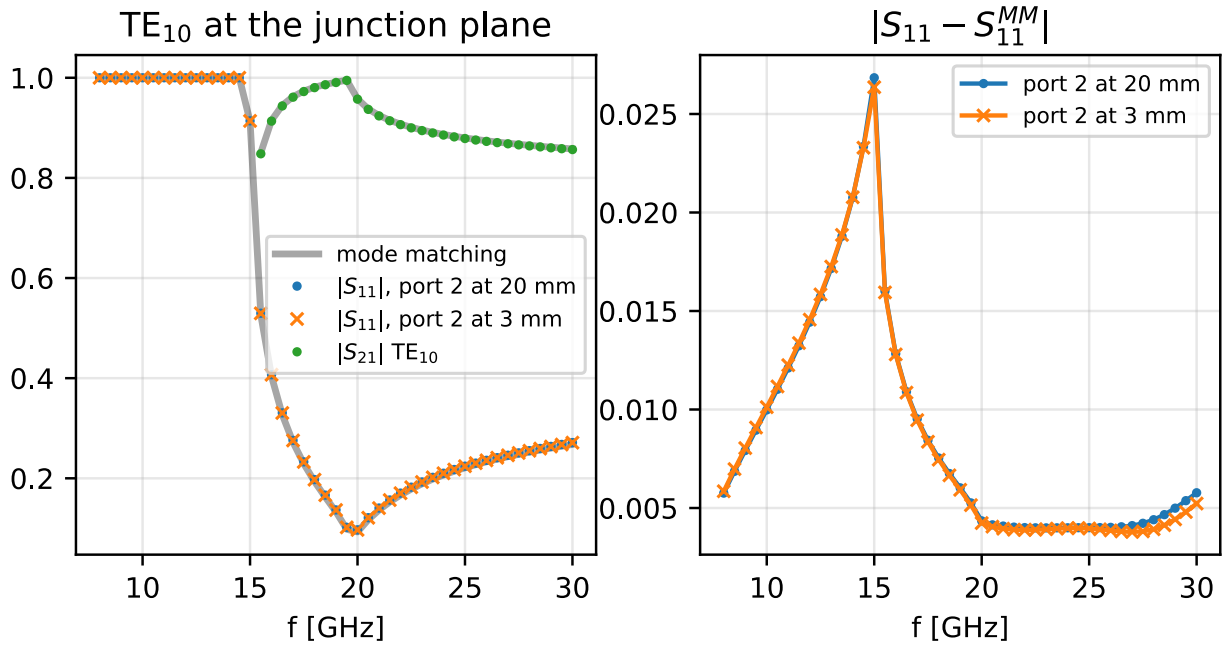


Figure 5: EMerge against mode matching. $|S_{11}|$ agrees to 0.004 and $\angle S_{11}$ to 1–2°, $|S_{21}|$ to 0.003. With the narrow port 3 mm from the step instead of 20 mm the result is the same to 10^{-3}: the evanescent modes are absorbed by the port, not reflected.

check	**result**
$\max\lvert S_{11} - S_{11}^{\mathrm{MM}}\rvert$ (port 2 at 20 / 3 mm)	0.027 / 0.026 (mean 0.009), largest at the 15 GHz cutoff
power over propagating modes, $\lvert\sum \lvert S_{m1}\rvert^2 - 1\rvert$	$2 \cdot 10^{-6}$
uniform dielectric guide, `ModalPort`, $\max\lvert S_{11}\rvert$	0.41 before, < 0.001 after the fix
evanescent transmission, 10 mm guide at 12 GHz	$\lvert S_{21}\rvert = e^{-\alpha L}$, no reflection

Table 2: Results of `dev/waveport_step.py`, `dev/modal_port_beta_bug.py` and `tests/test_wave_port.py`.

8 Floquet ports

A `FloquetPort` terminates a periodic unit cell (`RectCell`, `HexCell`) above and below. Its modes are known in closed form, so it needs no eigen solver and no tracking. It does need the rest of the machinery: several modes with different γ, a fixed mode set, and exact absorption of evanescent modes.

8.1 Floquet modes

Let $\boldsymbol{a}_1, \boldsymbol{a}_2$ be the lattice vectors of the cell and $\boldsymbol{b}_1, \boldsymbol{b}_2$ the reciprocal vectors, $\boldsymbol{b}_i \cdot \boldsymbol{a}_j = 2\pi\delta_{ij}$. The periodic boundaries impose the phase $e^{-j\boldsymbol{k}_{t,00}\cdot\boldsymbol{a}}$ with $\boldsymbol{k}_{t,00} = k_0 \sin\theta(\cos\varphi, \sin\varphi)$. Every field that satisfies these boundaries is a sum of Floquet harmonics

$$\boldsymbol{k}_{t,mn} = \boldsymbol{k}_{t,00} + m\boldsymbol{b}_1 + n\boldsymbol{b}_2, \quad k_{z,mn} = \sqrt{\varepsilon_r k_0^2 - |\boldsymbol{k}_{t,mn}|^2}, \quad \operatorname{Im} k_z \le 0, \tag{15}$$

each as a TE and a TM plane wave. With $\hat{\boldsymbol{t}} = \boldsymbol{k}_t / |\boldsymbol{k}_t|$ (the scan direction $(\cos\varphi, \sin\varphi)$ when $\boldsymbol{k}_t = 0$) the transverse fields are

$$\boldsymbol{e}_{\mathrm{TE}} = A_{\mathrm{TE}}(\hat{\boldsymbol{z}} \times \hat{\boldsymbol{t}})e^{-j\boldsymbol{k}_t\cdot\boldsymbol{r}}, \quad \boldsymbol{e}_{\mathrm{TM}} = -A_{\mathrm{TM}}\hat{\boldsymbol{t}}e^{-j\boldsymbol{k}_t\cdot\boldsymbol{r}}. \tag{16}$$

The amplitudes give 1 W through the cell area S for a propagating mode:

$$A_{\mathrm{TE}}^2 = \frac{2k_0 Z_0}{S\,|k_z|}, \quad A_{\mathrm{TM}}^2 = \frac{2Z_0\,|k_z|}{S k_0 \varepsilon_r}. \tag{17}$$

For an evanescent mode the same expressions with $|k_z|$ normalize the amplitude, as for the `WavePort` (Section 3). Plane waves have $\hat{\boldsymbol{n}} \times \nabla \times \boldsymbol{E} = \gamma \boldsymbol{E}_t$ with

$$\gamma_{\mathrm{TE}} = jk_z, \quad \gamma_{\mathrm{TM}} = j\varepsilon_r k_0^2 / k_z. \tag{18}$$

For an evanescent mode, $k_z = -j\alpha$ gives $\gamma_{\mathrm{TE}} = \alpha$ and $\gamma_{\mathrm{TM}} = -\varepsilon_r k_0^2/\alpha$.

Over one cell the harmonics are orthogonal in the **conjugate** product: $\iint_S \boldsymbol{e}_i \cdot \boldsymbol{e}_j^*\,\mathrm{d}S = N_i\delta_{ij}$. Two harmonics differ by $e^{-j(m\boldsymbol{b}_1+n\boldsymbol{b}_2)\cdot\boldsymbol{r}}$, which integrates to zero over a cell. The TE and TM modes of one harmonic are orthogonal vectors. The non-conjugate product of the `WavePort` (Section 3) does **not** work here: under an oblique scan $\boldsymbol{e}_i \cdot \boldsymbol{e}_i$ contains $e^{-2j\boldsymbol{k}_t\cdot\boldsymbol{r}}$, which is not constant over the cell.

8.2 Port operator

The port imposes $\hat{\boldsymbol{n}} \times \nabla \times \boldsymbol{E} = \Gamma \boldsymbol{E}_t$ with

$$\Gamma \boldsymbol{E}_t = \gamma_r \boldsymbol{E}_t + \sum_{i \in \mathcal{B}} (\gamma_i - \gamma_r)\boldsymbol{e}_i \frac{\iint \boldsymbol{E}_t \cdot \boldsymbol{e}_i^*\,\mathrm{d}S}{N_i}, \quad \gamma_r = jk_0\sqrt{\varepsilon_r}. \tag{19}$$

On the span of the basis $\mathcal{B}$, every mode is absorbed with its own γ_i, which is exact. The rest of the field sees a first-order absorbing boundary, γ_r. If $\mathcal{B}$ held every harmonic the mesh can resolve, Equation 19 would be the exact Dirichlet-to-Neumann map of the half-space. In the FEM system Equation 19 becomes

$$\gamma_r M + \sum_i \frac{\gamma_i - \gamma_r}{N_i} \boldsymbol{G}_i \boldsymbol{G}_i^H, \quad G_{i,k} = \int \boldsymbol{N}_k \cdot \boldsymbol{e}_i\,\mathrm{d}S, \tag{20}$$

a Robin mass matrix plus a dense Hermitian rank-1 term per basis mode on the port DoFs, the same block pattern as the dense WPBC. A port mode j is a combination $\sum_i c_{ji}\boldsymbol{e}_i$ of basis modes. Its excitation is $-2\sum_i c_{ji}\gamma_i \boldsymbol{G}_i$. Its outgoing amplitude is the conjugate projection

$$b_j = \frac{\iint (\boldsymbol{E} - a_j\boldsymbol{e}_j) \cdot \boldsymbol{e}_j^*\,\mathrm{d}S}{\iint |\boldsymbol{e}_j|^2\,\mathrm{d}S}. \tag{21}$$

With 1 W modes $|S_{ij}|^2$ is the diffraction efficiency of a propagating order.

8.3 Mode set

As for the `WavePort` (Section 5), the modes are fixed at the highest frequency, at the current scan angle:

- `number_of_modes=1` (the default): one S-matrix mode, the mix $\cos\psi\,\boldsymbol{e}_{\mathrm{TE},00} + \sin\psi\,\boldsymbol{e}_{\mathrm{TM},00}$ of the polarization angle ψ. This is the old single-mode port. TE_{00} and TM_{00} are both still in $\mathcal{B}$ and both absorbed exactly; before this change the port used one blended γ.
- `number_of_modes=2`: TE_{00} and TM_{00} as separate S-matrix modes (port modes $p.1$ and $p.2$).
- More: also the grating modes $(m, n) \neq (0, 0)$, TE before TM, ordered by $|\boldsymbol{k}_{t,mn}|^2$ at $f_{\max}$. The specular $(0, 0)$ harmonic always comes first.

- `None`: all modes that propagate at $f_{\max}$.

A grating mode in the set that is below cutoff at a lower frequency stays in the S-matrix as an evanescent mode ($k_z = -j\alpha$), the same as for the `WavePort`.

8.4 Validation

`dev/floquet_ports.py`, two Floquet ports on a cell:

1. **Uniform slab** ($\varepsilon_r = 4$, 6 mm), 10 GHz, 10 × 10 mm cell, $\theta = 0\ldots70°$ at $\varphi = 0°$ and 30°, TE_{00} and TM_{00}. The reference is the analytic slab response.
2. **Lamellar grating** (period 30 mm, 50 % bars of $\varepsilon_r = 4$, 6 mm thick), normal incidence, TE (E_y), 8.25–15.75 GHz. The $(\pm1, 0)$ orders propagate above $c/\Lambda = 10$ GHz. The reference is RCWA (Moharam & Gaylord, 81 orders; it reproduces the analytic slab to 10^{-15} and conserves energy to 10^{-14}).
3. **Hexagonal (brick) lattice** (20 × 12 mm, lattice vectors $(a, 0)$ and $(a/2, b)$) with an $\varepsilon_r = 6$ post, 20 GHz, scan $(20°, 30°)$. Six modes propagate. The ports sit 10 mm or 4 mm from the post.

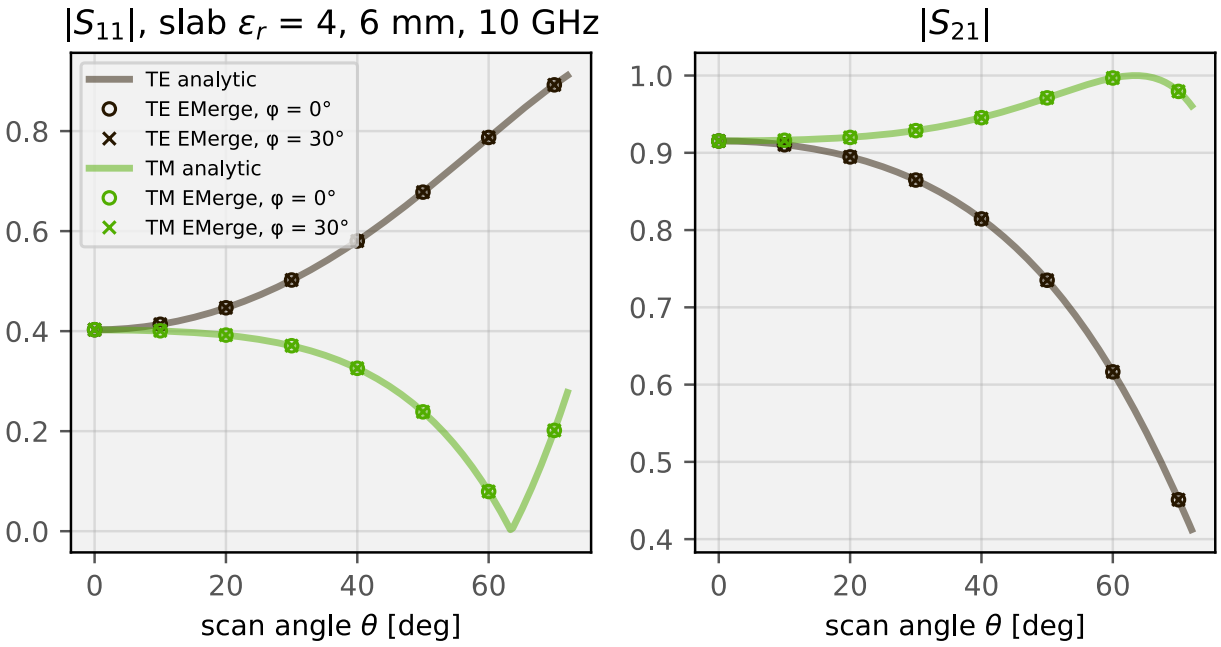


Figure 6: Uniform slab: TE and TM against the analytic response at both scan planes. The largest complex error in S_{11} and S_{21} is $4 \cdot 10^{-4}$, cross-polarization $< 2 \cdot 10^{-5}$, and power is conserved to $1.5 \cdot 10^{-4}$.

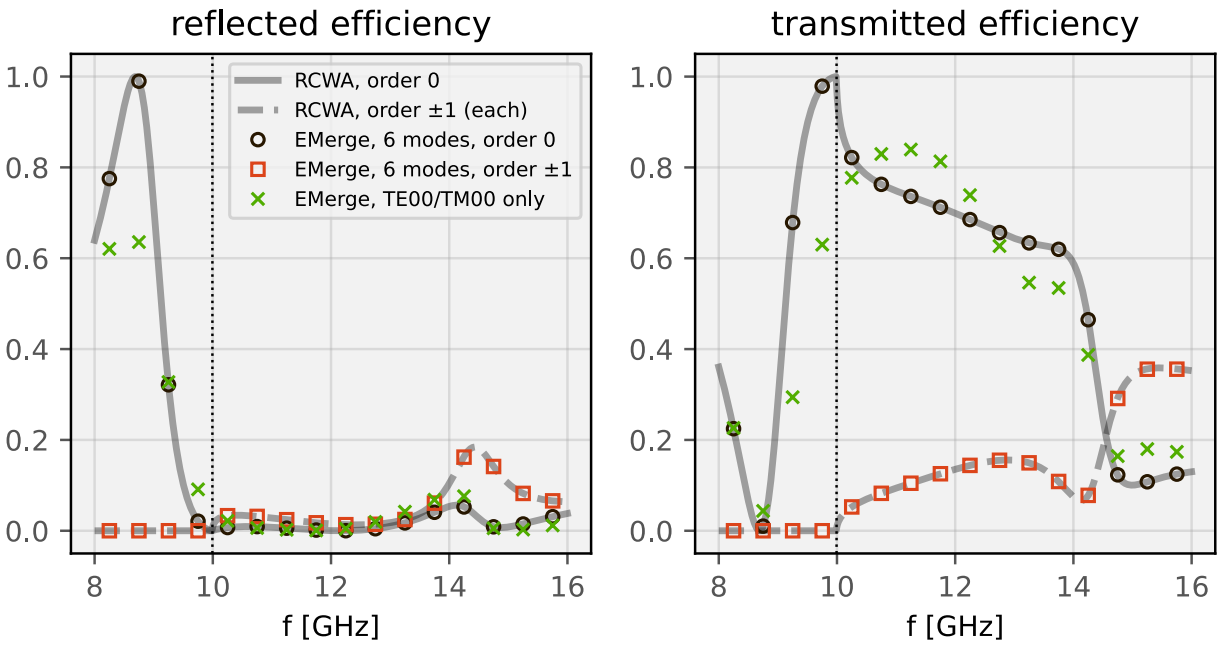


Figure 7: Lamellar grating against RCWA. With the six modes of 16 GHz in the ports (`number_of_modes=None`), every efficiency agrees to 0.0015 and power is conserved to $3 \cdot 10^{-3}$ (worst near the 14.5 GHz resonance, $< 10^{-4}$ elsewhere). With only $\mathrm{TE}_{00}/\mathrm{TM}_{00}$ the efficiencies are off by up to 0.38 and 15–83 % of the power is lost, also **below** 10 GHz. There the $(\pm 1, 0)$ orders are evanescent but, 12 mm from the grating, still at $e^{-1.4}$ of their amplitude. Terminated with γ_r instead of their own γ, they reflect.

modes per port	**port distance**	$\max\lvert\sum\lvert S\rvert^2 - 1\rvert$	S_{11} TE_{00}	S_{21} TE_{00}
6 (None)	10 mm	$1.7 \cdot 10^{-2}$	$-0.135 + 0.179j$	$-0.347 - 0.617j$
6 (None)	4 mm	$1.8 \cdot 10^{-1}$	$-0.128 + 0.115j$	$-0.333 - 0.535j$
14	10 mm	$3.4 \cdot 10^{-4}$	$-0.135 + 0.185j$	$-0.350 - 0.625j$
14	4 mm	$5.6 \cdot 10^{-3}$	$-0.133 + 0.185j$	$-0.350 - 0.624j$
30	10 mm	$3.2 \cdot 10^{-4}$	$-0.135 + 0.185j$	$-0.350 - 0.625j$
30	4 mm	$3.5 \cdot 10^{-4}$	$-0.134 + 0.185j$	$-0.350 - 0.625j$

Table 3: Hexagonal lattice with a post, S-parameters de-embedded to the post. With only the propagating modes in the port, the strong evanescent harmonics near the post are terminated by γ_r and the result depends on the port distance. With 30 modes (24 of them evanescent) the port 4 mm from the post gives the same S-parameters as at 10 mm, to $5 \cdot 10^{-4}$.

The hexagonal runs also exposed an old bug: a scan angle set with `set_scanangle` **before** the ports (or the periodic boundaries) were created never reached them. The ports then assumed normal incidence while the walls were phased for the scan, and $|S_{11}|$ of the slab came out 10 % low at 20°. The cell now stores the scan angle and applies it to boundaries and ports created later.

Choosing the number of modes. `None` is right when the ports are far from the structure compared with the decay length $1/\alpha$ of the first evanescent harmonics. Close to a strongly scattering structure, add evanescent harmonics until the S-parameters stop changing (Table 3). Each mode adds a right-hand side and its projections to the post-processing, but the dense block on the port face has the same size for any number of modes.

9 Limitations

- A mode exactly at cutoff at a sample frequency ($|\beta^2| < 10^{-4} k_0^2 \varepsilon_{r,\max} \mu_{r,\max}$) cannot be told apart from the spurious cluster. Tracking then stops with an error; moving the frequency point slightly avoids it.
- Degenerate pairs are orthogonalized in the power product only if they propagate. A degenerate evanescent pair ($\mathrm{TE}_{11}/\mathrm{TM}_{11}$ below cutoff) keeps the arbitrary basis of the eigen solver, and the tracker matches it by overlap.
- Evanescent modes have no characteristic impedance; their Z_0 is not defined.

- The bound Equation 12 is proven for homogeneous, TE-type and separable problems only.
- Floquet ports assume a homogeneous, non-magnetic medium at the port face (`er`, $\mu_r = 1$) and a face normal to z. A grating mode exactly at cutoff (a Wood/Rayleigh anomaly, $k_z = 0$) has an infinite γ_{TM}; it is clipped to $|k_z| = 10^{-6} k_0$ with a warning.
- Under an oblique scan the system is not reciprocal in the usual sense ($S_{ij} = S_{ji}$ holds for the reversed scan angle), so the reciprocity correction of the S-matrix should not be used with Floquet ports.